

UNIT - I

PRECISION FARMING

Q1. Ground Based Sensors

Ground Based Sensors are devices placed directly in the field to collect information about soil, crops, weather, and farm conditions. They are an important part of **Precision Agriculture**.

Sensor

A **sensor** is a device that detects changes in the environment such as **temperature, moisture, pressure, light, or movement** and converts them into electrical signals. These signals are used to monitor and control farming activities.

Types of Ground Based Sensors

1. Soil Moisture Sensor

- Measures moisture near plant roots.
- Helps schedule irrigation correctly.
- Prevents overwatering and underwatering.
- Improves plant growth.

2. Soil Temperature Sensor

- Measures real-time soil temperature.
- Helps predict soil-borne diseases.
- Used in agricultural research.
- Works well even in humid conditions.

3. Leaf Wetness Sensor

- Detects moisture on leaf surfaces.
- Helps identify fungal and bacterial diseases early.
- Supports pest and disease management.

4. Air Humidity Sensor

- Measures the moisture in the air.
- Helps predict disease and pest attacks.
- Improves crop management and reduces farming cost.

5. Wind Speed and Direction Sensor

- Measures wind speed and wind direction.
- Helps during pesticide spraying.
- Improves pollination and reduces crop damage.

6. Rainfall Sensor

- Measures the amount of rainfall.
- Helps farmers decide the correct irrigation time.
- Saves water and improves crop health.

7. Temperature Sensor

- Measures temperature around the crop canopy.
- Helps reduce disease and pest attack.
- Supports healthy crop growth.

Advantages of Ground Based Sensors

- Continuous monitoring of crops and soil.
- Saves water and fertilizers.
- Detects pests and diseases early.
- Increases crop yield and quality.
- Reduces labour and farming cost.
- Supports precision farming.
- Provides accurate real-time information.

Q2. Remote Sensing Applications in Agriculture

Introduction

Remote sensing is the process of collecting information about crops, soil, and water **without directly touching them**. It uses **satellites, drones, aircraft, and sensors** to collect images and data.

Major Applications of Remote Sensing

1. Crop Monitoring
 2. Yield Estimation
 3. Soil Analysis and Mapping
 4. Disease and Pest Detection
 5. Water Management
 6. Crop Health Monitoring
 7. Weather and Climate Monitoring
 8. Land Use and Crop Mapping
 9. Precision Agriculture
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1. Crop Monitoring

Remote sensing is used to **monitor crop growth and development**.

How it works:

- Satellite or drone cameras capture images of the farm.
 - The images are analyzed using computer software.
 - Different crop growth stages can be identified.
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2. Yield Estimation

Remote sensing helps to **estimate the amount of crop production before harvesting**.

How it works:

- Satellite or drone images are collected.
- Crop health and vegetation information are analyzed.
- This information is used to estimate expected crop yield.

Uses:

- Helps farmers plan harvesting.
 - Helps estimate future production.
 - Helps in storage planning.
 - Helps in market planning.
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3. Soil Analysis and Mapping

Remote sensing helps to study and map **soil properties** over a large area.

It can help identify:

- Soil moisture
 - Soil type
 - Soil fertility
 - Salinity
 - Nutrient conditions
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4. Disease and Pest Detection

Remote sensing can help detect **crop diseases and pest attacks** at an early stage.

How it works:

- Diseased plants reflect light differently from healthy plants.
- Satellite or drone sensors capture these differences.

- Software analyzes the images.
 - Problem areas are identified.
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5. Water Management

Remote sensing helps farmers manage **water and irrigation** efficiently.

How it works:

- Sensors and satellite images provide information about crop and soil moisture.
- Areas with low moisture can be identified.
- Irrigation can be provided only to the required areas.

Uses:

- Saves water.
- Prevents over-irrigation.
- Identifies dry areas.
- Helps plan irrigation schedules.

Example: If one part of a field is dry, irrigation can be provided mainly to that area.

6. Crop Health Monitoring

Remote sensing helps determine the **health of crops**. It measures using sensors for

- Detects plant stress.
- Identifies nutrient deficiency.
- Monitors plant growth.
- Helps improve crop management.

7. Weather and Climate Monitoring

Remote sensing provides information about **weather and climate conditions**.

It can help monitor:

- Temperature
 - Rainfall
 - Cloud cover
 - Humidity
 - Drought conditions
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8. Precision Agriculture

Remote sensing is an important part of **precision agriculture**.

It provides information about the exact condition of different areas of a farm.

Uses:

- Applies fertilizer only where required.
- Provides water based on crop needs.
- Detects disease early.
- Reduces input costs.
- Increases crop production.